

Лабораторная работа №1

Операционные системы

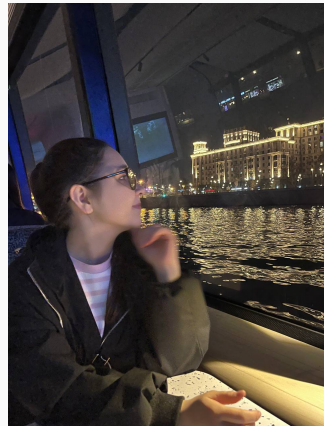
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15 июня 2026

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Информация

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Цель работы

- приобретение практических навыков установки операционной системы на виртуальную машину
- настройки минимально необходимых для дальнейшей работы сервисов

Задание

- освоить базовые команды терминала Linux
- получить системную информацию.

Выполнение лабораторной работы

Создание виртуальной машины

Создала новую виртуальную машину в графическом интерфейсе и указала имя виртуальной машины, соответствующее моему логину в дисплейном классе.

Указала размер основной памяти виртуальной машины — 4096 МБ.

Задала конфигурацию жёсткого диска — загрузочный, VDI.

Задала размер диска — 80 ГБ.

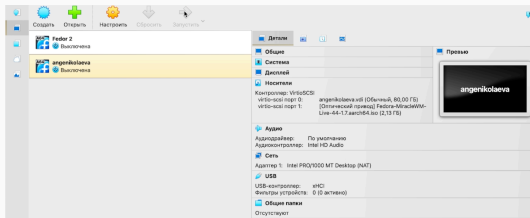


Рис. 1: Создание новой виртуальной машины в графическом интерфейсе

Подключение образа

Добавила новый привод оптических дисков и выбрала образ.



Рис. 2: Выбранный образ

Обновление пакетов

Обновила все пакеты и установила программу для удобства работы в консоли.

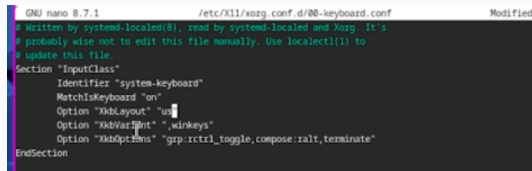
```
[147/155] Removing gettext-libs-0:0.19.1-5.fc44.aarch64 100% | 333.0 B/s | 9.0 B | 00m00s
[148/155] Removing gettext-envsubst-0:0.26-4.fc44.aarch64 100% | 333.0 B/s | 9.0 B | 00m00s
[149/155] Removing libomp-0:16.0.1-0.10.fc44.aarch64 100% | 600.0 B/s | 9.0 B | 00m00s
[150/155] Removing llvmfilesystem-0:22.1.1-2.fc44.aarch64 100% | 205.0 B/s | 7.0 B | 00m00s
[151/155] Removing glibc-0:2.43-2.fc44.aarch64 100% | 1.3 KiB/s | 101.0 B | 00m00s
[152/155] Removing glibc-all-langpacks-0:2.43-2.fc44.aarch64 100% | 2.3 KiB/s | 42.0 B | 00m00s
[153/155] Removing glibc-gconv-extra-0:2.43-2.fc44.aarch64 100% | 7.3 KiB/s | 632.0 B | 00m00s
[154/155] Removing glibc-common-0:2.43-2.fc44.aarch64 100% | 1.9 KiB/s | 53.0 B | 00m00s
[155/155] Removing libgcc-0:16.0.1-0.10.fc44.aarch64 100% | 4.0 B/s | 11.0 B | 00m02s
Complete!
root@fedora:~# dnf -y install dkms
Updating and loading repositories:
Repositories loaded.
Package Arch Version Repository Size
Installing:
dkms noarch 0:3.4.1-1.fc44 updates 333.9 KiB

Transaction Summary:
Installing: 1 package

Total size of inbound packages is 121 KiB. Need to download 121 KiB.
After this operation, 334 KiB extra will be used (install 334 KiB, remove 0 B)
[1/1] dkms-0:3.4.1-1.fc44.noarch 100% | 51.3 KiB/s | 121.4 KiB | 00m02s
-----
[1/1] Total 100% | 37.8 KiB/s | 121.4 KiB | 00m03s
Running transaction:
[1/3] Verify package files 100% | 76.0 B/s | 1.0 B | 00m00s
[2/3] Prepare transaction 100% | 7.0 B/s | 1.0 B | 00m00s
[3/3] Installing dkms-0:3.4.1-1.fc44.noarch 100% | 389.3 KiB/s | 337.5 KiB | 00m01s
>>> Running %post scriptlet: dkms-0:3.4.1-1.fc44.noarch
>>> Finished %post scriptlet: dkms-0:3.4.1-1.fc44.noarch
>>> Scriptlet output
>>> Created symlink '/etc/systemd/system/multi-user.target.wants/dkms.service' ->
'/usr/lib/systemd/system/dkms.service'.
>>>
Complete!
```

Отключение SELinux и настройка обновлений

Отключила систему безопасности SELinux.

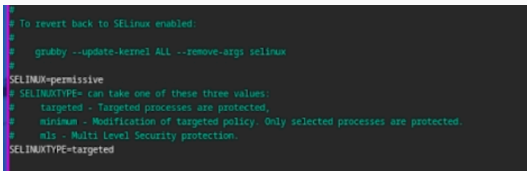


```
GNU nano 8.7.1 /etc/X11/xorg.conf.d/00-keyboard.conf Modified
# Written by systemd-locale(8), read by systemd-locale and Xorg. It's
# probably wise not to edit this file manually. Use localectl(1) to
# update this file.
Section "InputClass"
    Identifier "system-keyboard"
    MatchIsKeyboard "on"
    Option "XkbLayout" "us"
    Option "XkbVariant" ",winkeys"
    Option "XkbOptions" "grp:rctrl_toggle,compose:ralt,terminate"
EndSection
```

Рис. 4: Установка автоматического обновления

Настройка клавиатуры (Sway)

Запустила терминальный мультиплексор `tmux` и создала конфигурационный файл `~/.config/sway/config.d/95-system-keyboard-config.conf`.
Отредактировала конфигурационный файл.

A terminal window with a dark background and light-colored text. The text shows instructions for reverting SELinux to an enabled state using the 'grubby' command, followed by a list of SELinux policy types and their descriptions, and finally setting 'SELINUXTYPE=targeted'.

```
# To revert back to SELinux enabled:
#
# grubby --update-kernel ALL --remove-args selinux
#
SELINUX=permissive
# SELINUXTYPE= can take one of these three values:
#   targeted - Targeted processes are protected,
#   minimum - Modification of targeted policy. Only selected processes are protected.
#   mls - Multi Level Security protection.
SELINUXTYPE=targeted
```

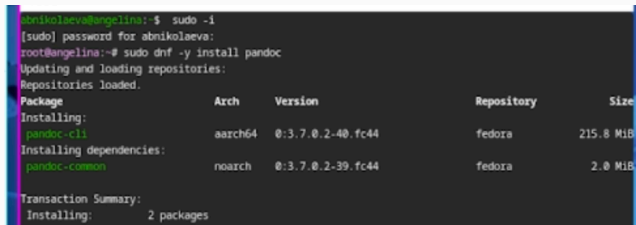
Рис. 5: Отредактированный конфигурационный файл

Настройка имени хоста и группы

Установила имя хоста, добавила своего пользователя в группу vboxsf.

```
abnikolaeva@fedora:~$ sudo -i
[sudo] password for abnikolaeva:
root@fedora:~# adduser -G wheel 1032253612
useradd: group 'wheel' does not exist
root@fedora:~# adduser -G wheel 1032253612
useradd: group 'wheel' does not exist
root@fedora:~# adduser -G wheel 1032253612
useradd: invalid user name '1032253612'
root@fedora:~# adduser -G wheel angelina
root@fedora:~# passwd angelina
New password:
Retype new password:
Sorry, passwords do not match.
passwd: Authentication token manipulation error
passwd: password unchanged
root@fedora:~# passwd angelina
New password:
Retype new password:
passwd: password updated successfully
root@fedora:~# hostnamectl set-hostname angelina
root@fedora:~# hostnamectl
  Static hostname: angelina
            Icon name: computer
        Machine ID: 935e0c6d505f48aaae81aa1faca4595b
         Boot ID: 3215cb4043cb42b8a5d31d70d8c79a33
  Operating System: Fedora Linux 44 (MiracleWM)
       CPE OS Name: cpe:/o:fedoraproject:fedora:44
        OS Support End: Wed 2027-05-19
 OS Support Remaining: 11month 4d
      Kernel Name: Linux 6.10.10-300.fc44.aarch64
```

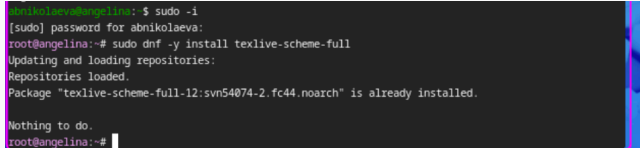
Запустила терминальный мультиплексор `tmux`, переключилась на роль супер-пользователя и установила средство `pandoc` для работы с языком разметки Markdown.

A terminal window showing the installation of pandoc. The user 'abnikolaeva' runs 'sudo -i' to become root. Then, 'sudo dnf -y install pandoc' is executed. The terminal shows the process of updating repositories and loading them. A table lists the packages to be installed: 'pandoc-cli' (aarch64, version 0:3.7.0.2-40.fc44, size 215.8 MiB) and 'pandoc-common' (noarch, version 0:3.7.0.2-39.fc44, size 2.0 MiB). A transaction summary shows that 2 packages will be installed.

```
abnikolaeva@angelina:~$ sudo -i
[sudo] password for abnikolaeva:
root@angelina:~# sudo dnf -y install pandoc
Updating and loading repositories:
Repositories loaded.
Package Arch Version Repository Size
Installing:
pandoc-cli aarch64 0:3.7.0.2-40.fc44 fedora 215.8 MiB
Installing dependencies:
pandoc-common noarch 0:3.7.0.2-39.fc44 fedora 2.0 MiB
Transaction Summary:
Installing: 2 packages
```

Рис. 7: Запуск `tmux` и установка `pandoc`

Установила дистрибутив TeXlive.

A terminal window with a dark background and light-colored text. The text shows a user named 'abnikolaeva' at a host named 'angelina' running 'sudo -i' to become root. Then, they run 'sudo dnf -y install texlive-scheme-full'. The terminal output shows that repositories are loaded and the package 'texlive-scheme-full-12:svn54074-2.fc44.noarch' is already installed, so nothing needs to be done.

```
abnikolaeva@angelina:~$ sudo -i
[sudo] password for abnikolaeva:
root@angelina:~# sudo dnf -y install texlive-scheme-full
Updating and loading repositories:
Repositories loaded.
Package "texlive-scheme-full-12:svn54074-2.fc44.noarch" is already installed.
Nothing to do.
root@angelina:~#
```

Рис. 8: Установка TeXlive

Домашнее задание

Получила версию ядра Linux.

Объём оперативной памяти

Получила объём доступной оперативной памяти.

```
root@angelina:~# dmesg | grep -i "mem"
[ 0.000000] efi: ACPI 2.0=0x0de13d018 ATTR=0xddec9018 MOKvar=0xdd500000 INITRD=0xddecb98 RESERVE=0xddecb618
[ 0.000000] NUMA: Faking a node at [mem 0x0000000000000000-0x00000000127fffff]
[ 0.000000] NODE_DATA(0) allocated [mem 0x122de8280-0x122dfefff]
[ 0.000000] DMA [mem 0x0000000000000000-0x00000000ffffffff]
[ 0.000000] Normal [mem 0x0000000010000000-0x00000000127fffff]
[ 0.000000] Early memory node ranges
[ 0.000000] node 0: [mem 0x0000000000000000-0x00000000dd4fffff]
[ 0.000000] node 0: [mem 0x00000000dd500000-0x00000000dd50fffff]
[ 0.000000] node 0: [mem 0x00000000dd510000-0x00000000de15fffff]
[ 0.000000] node 0: [mem 0x00000000de160000-0x00000000de35fffff]
[ 0.000000] node 0: [mem 0x00000000de360000-0x00000000de3bfffff]
[ 0.000000] node 0: [mem 0x00000000de3c0000-0x00000000de4dfffff]
[ 0.000000] node 0: [mem 0x00000000de4e0000-0x00000000de63fffff]
[ 0.000000] node 0: [mem 0x00000000de640000-0x00000000de65fffff]
[ 0.000000] node 0: [mem 0x00000000de660000-0x00000000df2bfffff]
[ 0.000000] node 0: [mem 0x00000000df2c0000-0x00000000df69fffff]
[ 0.000000] node 0: [mem 0x00000000df6a0000-0x00000000dfffffff]
[ 0.000000] node 0: [mem 0x0000000010000000-0x00000000127fffff]
[ 0.000000] Init setup node 0 [mem 0x0000000000000000-0x00000000127fffff]
[ 0.000000] software IO TLB: mapped [mem 0x00000000c5350000-0x00000000c9350000] (64MB)
[ 0.000000] mem auto-init: stack:all(zero), heap alloc:on, heap free:off
[ 0.023185] Memory: 3673528K/4194304K available (22976K kernel code, 6030K rodata, 19488K rodata, 14720K init, 10298K bss, 449172K reserved, 65536K cma-reserved)
[ 0.030995] HugeTLB: 0 KiB vmmap can be freed for a 1.00 GiB page
[ 0.030996] HugeTLB: 0 KiB vmmap can be freed for a 32.0 MiB page
[ 0.030996] HugeTLB: 0 KiB vmmap can be freed for a 3.00 MiB page
```

Получила тип обнаруженного гипервизора и последовательность монтирования файловых систем.

```
root@angelina:~# dmesg | grep -i "Hypervisor detected"
root@angelina:~# dmesg | grep -i "Hypervisor"
root@angelina:~# dmesg | grep -i "root"
[ 0.000000] Kernel command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-0-rescue-935e0c6d505f48aaae81aaf
a4595b root=UUID=0447c779-3207-4485-9b10-a26e78f0b828 ro rootflags=subvol=root rhgb quiet
[ 0.000000] Root IRQ handler: gic_handle_irq
[ 0.035342] ACPI: PCI Root Bridge [PCI0] (domain 0000 [bus 00-0f])
[ 0.035414] pci_bus 0000:00: root bus resource [io 0x0000-0xffff window]
[ 0.035414] pci_bus 0000:00: root bus resource [mem 0xe0000000-0xffffffff window]
[ 0.035415] pci_bus 0000:00: root bus resource [mem 0xfcd20000-0xfcd2ffff window]
[ 0.035416] pci_bus 0000:00: root bus resource [bus 00-0f]
[ 0.035416] PCI: OF: of_root node is NULL, cannot create PCI host bridge node
[ 0.060162] Trying to unpack rootfs image as initramfs...
[ 2.899047] systemd[1]: initrd-switch-root.service: Deactivated successfully.
[ 2.899123] systemd[1]: Stopped initrd-switch-root.service - Switch Root.
[ 2.900993] systemd[1]: Stopped target initrd-switch-root.target - Switch Root.
[ 2.901063] systemd[1]: Stopped target initrd-root-fs.target - Initrd Root File System.
[ 2.922270] systemd[1]: plymouth-switch-root.service: Deactivated successfully.
[ 2.922290] systemd[1]: Stopped plymouth-switch-root.service - Plymouth switch root service.
[ 2.922457] systemd[1]: systemd-fsck-root.service: Deactivated successfully.
[ 2.923571] systemd[1]: Stopped systemd-fsck-root.service - File System Check on Root Device
```

Рис. 11: Получение последовательности монтирования файловых систем

Выводы

В ходе работы были освоены базовые приёмы работы в терминале Linux и оформления результата.